• **WHEELS—ROAD FORCE MEASUREMENT EQUIPMENT HUNTER GSP9700—SERVICE TIPS** • **VIBRATION—ROAD FORCE MEASUREMENT EQUIPMENT HUNTER GSP9700—SERVICE TIPS**	**Article No. 02-1-6**

FORD: 1997 THUNDERBIRD
1997-2000 CONTOUR
1997-2002 CROWN VICTORIA, ESCORT, MUSTANG, TAURUS
2000-2002 FOCUS
2002 THUNDERBIRD
1997-2002 E SERIES, EXPEDITION, EXPLORER, F-150, RANGER, WINDSTAR
1999-2002 SUPER DUTY F SERIES
2000-2002 EXCURSION
2001-2002 ESCAPE, EXPLORER SPORT TRAC, EXPLORER SPORT

LINCOLN: 1997-1998 MARK VIII
1997-2002 CONTINENTAL, TOWN CAR
2000-2002 LS
1998-2002 NAVIGATOR
2002 BLACKWOOD

MERCURY: 1997 COUGAR
1997-2000 MYSTIQUE
1997-2002 GRAND MARQUIS, SABLE
1999-2002 COUGAR
1997-2002 MOUNTAINEER, VILLAGER

ISSUE

The purpose of this bulletin is to provide guidance when using tire/wheel assembly road force measurement equipment such as the Hunter GSP9700. This bulletin is authored and endorsed by Hunter, Michelin, Goodyear, Continental/General, Hankook, Bridgestone/Firestone and Pirelli. Items addressed are software upgrades, road test (exercise tires), assembly center verification and warranty restrictions.

This type of equipment can be a valuable tool to improve customer satisfaction when used as a vibration repair tool. This type of equipment can also provide a high level of customer satisfaction when used as an assembly tool when the customer has purchased new wheels/tires in the "Around-the-Wheel" program.

ACTION

The Hunter GSP9700/9712 road force measurement tool is NOT to be used for determining if a tire or wheel meets a specification. It can be used as a service diagnostic tool to minimize road force variation of the tire/wheel assembly. The tool such as the Hunter GSP9700 can approximate the low point of the wheel and the high point of the tire. This allows the tire to be matched mounted to the wheel for lowest tire/wheel assembly road force variation. This will minimize the contribution to vibration from the tire/wheel assembly.

SERVICE INFORMATION

Tire/Wheel Assembly Road Force Measurement Tool Procedure:

NOTE
TRADITIONAL WHEEL BALANCER "MATCH-MOUNTING" OR "WEIGHT OPTIMIZATION" ACHIEVES WEIGHT MINIMIZATION AND WILL NOT ACHIEVE THE SAME RESULTS.

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NOTE
WARRANTY CLAIMS CANNOT BE SUBMITTED FOR TIRE/WHEEL REPLACEMENT BASED ON HUNTER GSP9700 RESULTS.

NOTE
UPGRADING SOFTWARE TO VERSION 2.1 OR GREATER WILL ENHANCE THE TECHNICIANS CAPABILITY TO MINIMIZE THE ROAD FORCE VARIATION OF THE TIRE/WHEEL ASSEMBLY. THE VERSION OF SOFTWARE CAN BE IDENTIFIED FROM THE START-UP SCREEN BY PRESSING THE CENTER SHIFT KEY ONCE (I.E. HAS TWO GREEN ARROWS) AND SELECTING THE K1 KEY ("IDENTIFY SOFTWARE"). IF THE SYSTEM IS IN THE BALANCE MODE, THE START-UP SCREEN CAN BE ACCESSED BY PRESSING THE "R" KEY TWICE (THIS IS JUST TO THE LEFT OF THE START KEY). CONTACT YOUR LOCAL HUNTER ENGINEERING REPRESENTATIVE TO ORDER SOFTWARE UPGRADES.

1. Check the inflation pressure of all four tires and adjust to the vehicle manufacturers recommended pressure. The vehicle manufacturers recommended tire pressure for the original equipment tires can be found on the drivers side door placard.

2. Road test the vehicle on a smooth road for a minimum of 10 miles. Performance of the road test at highway speeds is preferred.

NOTE
THIS MUST BE PERFORMED TO "EXERCISE" THE TIRES AND ELIMINATE "COLD" FLAT SPOTTING THAT OCCURS DUE TO SHORT-PERIOD VEHICLE STORAGE OR PARKING. IF THE VEHICLE'S TIRES ARE NOT EXERCISED FIRST, ACCURACY OF THE ROAD FORCE MEASUREMENTS CAN VARY SIGNIFICANTLY, EVEN IF THE VEHICLE IS PARKED/STORED IN ONE LOCATION FOR ONLY A SHORT PERIOD OF TIME (E.G. 20 MINUTES).

3. Perform the road force measurement IMMEDIATELY after exercising the tires. If the road force measurement cannot be performed immediately, lift all four tires off the ground with a hoist or jack stands to prevent measurement errors due to further flat spotting.

4. Assembly centering verification MUST be performed and PASSED prior to any road force measurement.

For those with a software version other than 2.1 the centering verification must be performed manually.

   a. Spin assembly to measure. Record the assembly R1H.

   b. Loosen wing-nut.

   c. Rotate and reposition assembly and cone/adapter to a different mounting location (180 degrees).

   d. Tighten wing-nut.

   e. Re-spin to measure. Record the assembly R1H.

   f. Compare results obtained in STEP a with those obtained in STEP e.

   g. If the difference in assembly R1H measurements is less than 5 lbs., the assembly centering verification PASSES and technician can proceed to the measurement/repair process. If the centering verification FAILS proceed to STEP h.

   h. The following items may contribute to an assembly centering verification FAILURE.
      - Cone/adapter does not fit properly or is incorrect for the application
      - The wheel mounting surface is damaged

5. For those with software version 2.1 the following procedure will lead the technician through a Centering Check.

   a. From the balance mode screen press the center SHIFT KEY ONCE (i.e. has two green arrows).

   b. Push the "K1" button to select Perform Centering Check.

   c. Follow the instructions.

   d. If Centering Check PASSES proceed to the measurement/repair process. If the Centering Check fails proceed to step e.

   e. The following items may contribute to an assembly centering verification FAILURE.
      - Cone/adapter does not fit properly or is incorrect for the application
      - The wheel mounting surface is damaged

6. Prior to installing assembly on the vehicle, balance assembly utilizing the two-plane dynamic balancing mode.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 306000, 703000